

9.17 The log-likelihood function for a mixture of Bernoulli distributions is given by (9.51):

$$\ln p(\mathbf{X}|\boldsymbol{\mu}, \boldsymbol{\pi}) = \sum_{n=1}^N \ln \left\{ \sum_{k=1}^K \pi_k p(\mathbf{x}_n|\boldsymbol{\mu}_k) \right\}$$

We have an upper bound on the term inside the natural log:

$$\sum_{k=1}^K \pi_k p(\mathbf{x}_n|\boldsymbol{\mu}_k) \leq \sum_{k=1}^K \pi_k = 1$$

since the probability mass function $p(\mathbf{x}_n|\boldsymbol{\mu}_k)$ of a discrete variable can never exceed 1.

This makes the upper bound on the log-likelihood to be 0, and it cannot have any singularities for which the likelihood goes to infinity.

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